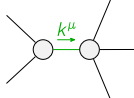
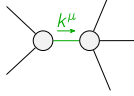
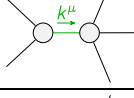
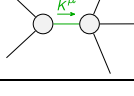
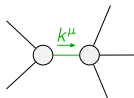
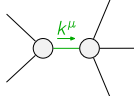
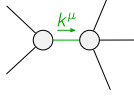
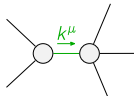


	$\mathcal{K}_0^{\#1} \uparrow$	$\mathcal{K}_0^{\#1} \uparrow^{\alpha\beta\chi}$				
	0	0				
$\mathcal{K}_0^{\#1} \uparrow^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1} \uparrow_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2} \uparrow_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1} \uparrow_{\alpha}$	$\mathcal{K}_{1^+}^{\#2} \uparrow_{\alpha}$
	$\mathcal{K}_{1^+}^{\#1} \uparrow^{\alpha\beta}$	0	0	0	0	0
	$\mathcal{K}_{1^+}^{\#2} \uparrow^{\alpha\beta}$	0	0	0	0	0
	$\mathcal{K}_{1^+}^{\#1} \uparrow^{\alpha}$	0	0	$\frac{Y_1 k^2}{2}$	$\frac{Y_2 k^2}{2\sqrt{2}}$	
	$\mathcal{K}_{1^+}^{\#2} \uparrow^{\alpha}$	0	0	$\frac{Y_2 k^2}{2\sqrt{2}}$	$\frac{Y_3 k^2}{4}$	$\mathcal{K}_{2^+}^{\#1} \uparrow_{\alpha\beta}$ $\mathcal{K}_{2^+}^{\#1} \uparrow_{\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1} \uparrow^{\alpha\beta}$	$\frac{3Y_4 k^2}{2}$	0	
			$\mathcal{K}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi}$	0	$\frac{3k^2 \kappa_{15}^{(4)}}{2}$	

	$\mathcal{J}_0^{\#1} \uparrow$	$\mathcal{J}_0^{\#1} \uparrow^{\alpha\beta\chi}$				
	0	0				
$\mathcal{J}_0^{\#1} \uparrow^{\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{\#1} \uparrow_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2} \uparrow_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#1} \uparrow_{\alpha}$	$\mathcal{J}_{1^+}^{\#2} \uparrow_{\alpha}$
	$\mathcal{J}_{1^+}^{\#1} \uparrow^{\alpha\beta}$	0	0	0	0	0
	$\mathcal{J}_{1^+}^{\#2} \uparrow^{\alpha\beta}$	0	0	0	0	0
	$\mathcal{J}_{1^+}^{\#1} \uparrow^{\alpha}$	0	0	$\frac{Y_3 k^2}{4 \text{Det}(1^+)}$	$-\frac{Y_2 k^2}{2\sqrt{2} \text{Det}(1^+)}$	
	$\mathcal{J}_{1^+}^{\#2} \uparrow^{\alpha}$	0	0	$-\frac{Y_2 k^2}{2\sqrt{2} \text{Det}(1^+)}$	$\frac{Y_1 k^2}{2 \text{Det}(1^+)}$	$\mathcal{J}_{2^+}^{\#1} \uparrow_{\alpha\beta}$ $\mathcal{J}_{2^+}^{\#1} \uparrow_{\alpha\beta\chi}$
			$\mathcal{J}_{2^+}^{\#1} \uparrow^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0	
			$\mathcal{J}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi}$	0	$\frac{2}{3k^2 \kappa_{15}^{(4)}}$	

Abbreviations used in matrices
$Y_1 == 3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}$ && $Y_2 == 2 \kappa_7^{(4)} + \kappa_{15}^{(4)}$ && $Y_3 == 3 \kappa_1^{(4)} - 6 \kappa_{15}^{(4)} + \kappa_2^{(4)} - 2 \kappa_7^{(4)}$ && $Y_4 == \kappa_1^{(4)} + \kappa_2^{(4)}$ &&
$\text{Det}(1^+) == \frac{1}{8} k^4 (9 \kappa_1^{(4)} \kappa_{15}^{(4)} - 18 \kappa_{15}^{(4)2} - 6 \kappa_1^{(4)} \kappa_2^{(4)} + 15 \kappa_{15}^{(4)} \kappa_2^{(4)} - 6 \kappa_2^{(4)2} - 6 \kappa_{15}^{(4)} \kappa_7^{(4)} - \kappa_7^{(4)2})$ &&
$\text{Det}(2^+) == \frac{3}{2} k^2 (\kappa_1^{(4)} + \kappa_2^{(4)})$

Lagrangian			
$\kappa_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \kappa_2 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta - \frac{3}{2} \kappa_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta +$ $2 \kappa_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta - \frac{3}{2} \kappa_2 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + 2 \kappa_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta -$ $\frac{3}{2} \kappa_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + \kappa_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{3}{2} \kappa_2 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta +$ $\kappa_7 \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \kappa_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \kappa_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \kappa_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_0^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha{}^\beta == 0$	
$\mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{\kappa_1^{(4)} + \kappa_{15}^{(4)} + \kappa_2^{(4)}}{\kappa_1^{(4)} \kappa_{15}^{(4)} + \kappa_{15}^{(4)} \kappa_2^{(4)}}$
	1	0	$-\frac{\kappa_1^{(4)} + \kappa_{15}^{(4)} + \kappa_2^{(4)}}{\kappa_1^{(4)} \kappa_{15}^{(4)} + \kappa_{15}^{(4)} \kappa_2^{(4)}}$
	1	0	$\frac{\sqrt{3 \kappa_1^{(4)2} + \kappa_{15}^{(4)2} + 2 \kappa_{15}^{(4)} \kappa_2^{(4)} + 3 \kappa_2^{(4)2} + 2 \kappa_1^{(4)} (\kappa_{15}^{(4)} + 3 \kappa_2^{(4)})}}{\kappa_{15}^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)})}$
	1	0	$-\frac{\sqrt{3 \kappa_1^{(4)2} + \kappa_{15}^{(4)2} + 2 \kappa_{15}^{(4)} \kappa_2^{(4)} + 3 \kappa_2^{(4)2} + 2 \kappa_1^{(4)} (\kappa_{15}^{(4)} + 3 \kappa_2^{(4)})}}{\kappa_{15}^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)})}$
	1	0	$-((-24 \kappa_1^{(4)} \kappa_{15}^{(4)} + 24 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_2^{(4)} - 24 \kappa_{15}^{(4)2} \kappa_2^{(4)} + 24 \kappa_{15}^{(4)} \kappa_2^{(4)2} + 4 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_7^{(4)} + 4 \kappa_{15}^{(4)} \kappa_2^{(4)} \kappa_7^{(4)} +$ $\sqrt{(\kappa_{15}^{(4)2} (9 \kappa_1^{(4)} + 324 \kappa_{15}^{(4)} + 1161 \kappa_2^{(4)} + 372 \kappa_2^{(4)3} \kappa_7^{(4)} + 30 \kappa_2^{(4)2} \kappa_7^{(4)2} + 4 \kappa_2^{(4)} \kappa_7^{(4)3} + \kappa_7^{(4)4} -$ $6 \kappa_1^{(4)3} (9 \kappa_{15}^{(4)} - 6 \kappa_2^{(4)} + 2 \kappa_7^{(4)}) + 108 \kappa_{15}^{(4)3} (\kappa_2^{(4)} + 2 \kappa_7^{(4)}) + 9 \kappa_{15}^{(4)2} (117 \kappa_2^{(4)2} + 12 \kappa_2^{(4)} \kappa_7^{(4)} + 8 \kappa_7^{(4)2}) +$ $3 \kappa_1^{(4)2} (375 \kappa_{15}^{(4)} - 810 \kappa_{15}^{(4)} \kappa_2^{(4)} + 402 \kappa_2^{(4)2} - 128 \kappa_{15}^{(4)} \kappa_7^{(4)} + 116 \kappa_2^{(4)} \kappa_7^{(4)} + 10 \kappa_7^{(4)2}) -$ $6 \kappa_{15}^{(4)} (387 \kappa_2^{(4)3} + 68 \kappa_2^{(4)2} \kappa_7^{(4)} - 5 \kappa_2^{(4)} \kappa_7^{(4)2} - 2 \kappa_7^{(4)3}) + 2 \kappa_1^{(4)} (162 \kappa_{15}^{(4)3} + 9 \kappa_{15}^{(4)2} (119 \kappa_2^{(4)} + 10 \kappa_7^{(4)}) -$ $3 \kappa_{15}^{(4)} (783 \kappa_2^{(4)2} + 132 \kappa_2^{(4)} \kappa_7^{(4)} - 7 \kappa_7^{(4)2}) + 2 (585 \kappa_2^{(4)3} + 183 \kappa_2^{(4)2} \kappa_7^{(4)} + 15 \kappa_2^{(4)} \kappa_7^{(4)2} + \kappa_7^{(4)3})))) /$ $(\kappa_{15}^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (-9 \kappa_1^{(4)} \kappa_{15}^{(4)} + 18 \kappa_{15}^{(4)2} + 6 \kappa_1^{(4)} \kappa_2^{(4)} - 15 \kappa_{15}^{(4)} \kappa_2^{(4)} + 6 \kappa_2^{(4)2} + 6 \kappa_{15}^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2}))$
	1	0	$(24 \kappa_1^{(4)} \kappa_{15}^{(4)} - 24 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_2^{(4)} + 24 \kappa_{15}^{(4)2} \kappa_2^{(4)} - 24 \kappa_{15}^{(4)} \kappa_2^{(4)2} - 4 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_7^{(4)} - 4 \kappa_{15}^{(4)} \kappa_2^{(4)} \kappa_7^{(4)} +$ $\sqrt{(\kappa_{15}^{(4)2} (9 \kappa_1^{(4)} + 324 \kappa_{15}^{(4)} + 1161 \kappa_2^{(4)} + 372 \kappa_2^{(4)3} \kappa_7^{(4)} + 30 \kappa_2^{(4)2} \kappa_7^{(4)2} + 4 \kappa_2^{(4)} \kappa_7^{(4)3} + \kappa_7^{(4)4} -$ $6 \kappa_1^{(4)3} (9 \kappa_{15}^{(4)} - 6 \kappa_2^{(4)} + 2 \kappa_7^{(4)}) + 108 \kappa_{15}^{(4)3} (\kappa_2^{(4)} + 2 \kappa_7^{(4)}) + 9 \kappa_{15}^{(4)2} (117 \kappa_2^{(4)2} + 12 \kappa_2^{(4)} \kappa_7^{(4)} + 8 \kappa_7^{(4)2}) +$ $3 \kappa_1^{(4)2} (375 \kappa_{15}^{(4)} - 810 \kappa_{15}^{(4)} \kappa_2^{(4)} + 402 \kappa_2^{(4)2} - 128 \kappa_{15}^{(4)} \kappa_7^{(4)} + 116 \kappa_2^{(4)} \kappa_7^{(4)} + 10 \kappa_7^{(4)2}) -$ $6 \kappa_{15}^{(4)} (387 \kappa_2^{(4)3} + 68 \kappa_2^{(4)2} \kappa_7^{(4)} - 5 \kappa_2^{(4)} \kappa_7^{(4)2} - 2 \kappa_7^{(4)3}) + 2 \kappa_1^{(4)} (162 \kappa_{15}^{(4)3} + 9 \kappa_{15}^{(4)2} (119 \kappa_2^{(4)} + 10 \kappa_7^{(4)}) -$ $3 \kappa_{15}^{(4)} (783 \kappa_2^{(4)2} + 132 \kappa_2^{(4)} \kappa_7^{(4)} - 7 \kappa_7^{(4)2}) + 2 (585 \kappa_2^{(4)3} + 183 \kappa_2^{(4)2} \kappa_7^{(4)} + 15 \kappa_2^{(4)} \kappa_7^{(4)2} + \kappa_7^{(4)3})))) /$ $(\kappa_{15}^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (-9 \kappa_1^{(4)} \kappa_{15}^{(4)} + 18 \kappa_{15}^{(4)2} + 6 \kappa_1^{(4)} \kappa_2^{(4)} - 15 \kappa_{15}^{(4)} \kappa_2^{(4)} + 6 \kappa_2^{(4)2} + 6 \kappa_{15}^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2}))$
	2	0	$-((-54 \kappa_1^{(4)2} \kappa_{15}^{(4)} + 36 \kappa_1^{(4)} \kappa_{15}^{(4)2} - 72 \kappa_{15}^{(4)3} + 18 \kappa_1^{(4)} \kappa_2^{(4)} -$ $108 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_2^{(4)} + 60 \kappa_{15}^{(4)2} \kappa_2^{(4)} + 36 \kappa_1^{(4)} \kappa_2^{(4)2} - 54 \kappa_{15}^{(4)} \kappa_2^{(4)2} + 18 \kappa_2^{(4)3} + 12 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_7^{(4)} -$ $24 \kappa_{15}^{(4)2} \kappa_7^{(4)} + 12 \kappa_{15}^{(4)} \kappa_2^{(4)} \kappa_7^{(4)} + 3 \kappa_1^{(4)} \kappa_7^{(4)2} - 4 \kappa_{15}^{(4)} \kappa_7^{(4)2} + 3 \kappa_2^{(4)} \kappa_7^{(4)2} + \sqrt{(5184 \kappa_{15}^{(4)6} +$ $108 \kappa_1^{(4)4} (3 \kappa_{15}^{(4)2} - 2 \kappa_{15}^{(4)} \kappa_2^{(4)} + 3 \kappa_2^{(4)2}) + 1728 \kappa_{15}^{(4)5} (\kappa_2^{(4)} + 2 \kappa_7^{(4)}) + 9 \kappa_2^{(4)2} (6 \kappa_2^{(4)2} + \kappa_7^{(4)2})^2 +$ $144 \kappa_{15}^{(4)4} (19 \kappa_2^{(4)2} + 16 \kappa_2^{(4)} \kappa_7^{(4)} + 8 \kappa_7^{(4)2}) - 48 \kappa_{15}^{(4)3} (9 \kappa_2^{(4)3} - 24 \kappa_2^{(4)2} \kappa_7^{(4)} - 23 \kappa_2^{(4)} \kappa_7^{(4)2} - 4 \kappa_7^{(4)3}) -$ $12 \kappa_{15}^{(4)} \kappa_2^{(4)} (18 \kappa_2^{(4)4} - 36 \kappa_2^{(4)3} \kappa_7^{(4)} - 9 \kappa_2^{(4)2} \kappa_7^{(4)2} - 6 \kappa_2^{(4)} \kappa_7^{(4)3} - 2 \kappa_7^{(4)4}) +$ $4 \kappa_{15}^{(4)2} (189 \kappa_2^{(4)4} + 180 \kappa_2^{(4)3} \kappa_7^{(4)} + 54 \kappa_2^{(4)2} \kappa_7^{(4)2} + 60 \kappa_2^{(4)} \kappa_7^{(4)3} + 4 \kappa_7^{(4)4}) +$ $36 \kappa_1^{(4)3} (36 \kappa_{15}^{(4)3} + 12 \kappa_{15}^{(4)2} (2 \kappa_2^{(4)} + \kappa_7^{(4)}) + 3 \kappa_2^{(4)} (12 \kappa_2^{(4)2} + \kappa_7^{(4)2}) - \kappa_{15}^{(4)} (24 \kappa_2^{(4)2} - 12 \kappa_2^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2})) +$ $9 \kappa_1^{(4)2} (432 \kappa_{15}^{(4)4} + 216 \kappa_2^{(4)4} + 36 \kappa_2^{(4)2} \kappa_7^{(4)2} + \kappa_7^{(4)4} + 48 \kappa_{15}^{(4)3} (5 \kappa_2^{(4)} + 4 \kappa_7^{(4)}) +$ $8 \kappa_{15}^{(4)2} (21 \kappa_2^{(4)2} + 22 \kappa_2^{(4)} \kappa_7^{(4)} + 5 \kappa_7^{(4)2}) + 4 \kappa_{15}^{(4)} (-36 \kappa_2^{(4)3} + 36 \kappa_2^{(4)2} \kappa_7^{(4)} + \kappa_2^{(4)} \kappa_7^{(4)2} + 2 \kappa_7^{(4)3}) +$ $6 \kappa_1^{(4)} (864 \kappa_{15}^{(4)5} + 144 \kappa_{15}^{(4)4} (7 \kappa_2^{(4)} + 4 \kappa_7^{(4)}) + 24 \kappa_{15}^{(4)3} (3 \kappa_2^{(4)2} + 20 \kappa_2^{(4)} \kappa_7^{(4)} + 9 \kappa_7^{(4)2}) +$ $8 \kappa_{15}^{(4)2} (36 \kappa_2^{(4)3} + 39 \kappa_2^{(4)2} \kappa_7^{(4)} + 12 \kappa_2^{(4)} \kappa_7^{(4)2} + 5 \kappa_7^{(4)3}) + 3 \kappa_2^{(4)} (72 \kappa_2^{(4)4} + 18 \kappa_2^{(4)2} \kappa_7^{(4)2} + \kappa_7^{(4)4}) +$ $\kappa_{15}^{(4)} (-144 \kappa_2^{(4)4} + 216 \kappa_2^{(4)3} \kappa_7^{(4)} + 30 \kappa_2^{(4)2} \kappa_7^{(4)2} + 24 \kappa_2^{(4)} \kappa_7^{(4)3} + 4 \kappa_7^{(4)4})))) /$ $(\kappa_{15}^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (-9 \kappa_1^{(4)} \kappa_{15}^{(4)} + 18 \kappa_{15}^{(4)2} + 6 \kappa_1^{(4)} \kappa_2^{(4)} - 15 \kappa_{15}^{(4)} \kappa_2^{(4)} + 6 \kappa_2^{(4)2} + 6 \kappa_{15}^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2}))$
	2	0	$(54 \kappa_1^{(4)2} \kappa_{15}^{(4)} - 36 \kappa_1^{(4)} \kappa_{15}^{(4)2} + 72 \kappa_{15}^{(4)3} - 18 \kappa_1^{(4)} \kappa_2^{(4)} +$ $108 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_2^{(4)} - 60 \kappa_{15}^{(4)2} \kappa_2^{(4)} - 36 \kappa_1^{(4)} \kappa_2^{(4)2} + 54 \kappa_{15}^{(4)} \kappa_2^{(4)2} - 18 \kappa_2^{(4)3} - 12 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_7^{(4)} +$ $24 \kappa_{15}^{(4)2} \kappa_7^{(4)} - 12 \kappa_{15}^{(4)} \kappa_2^{(4)} \kappa_7^{(4)} - 3 \kappa_1^{(4)} \kappa_7^{(4)2} + 4 \kappa_{15}^{(4)} \kappa_7^{(4)2} - 3 \kappa_2^{(4)} \kappa_7^{(4)2} + \sqrt{(5184 \kappa_{15}^{(4)6} +$ $108 \kappa_1^{(4)4} (3 \kappa_{15}^{(4)2} - 2 \kappa_{15}^{(4)} \kappa_2^{(4)} + 3 \kappa_2^{(4)2}) + 1728 \kappa_{15}^{(4)5} (\kappa_2^{(4)} + 2 \kappa_7^{(4)}) + 9 \kappa_2^{(4)2} (6 \kappa_2^{(4)2} + \kappa_7^{(4)2})^2 +$ $144 \kappa_{15}^{(4)4} (19 \kappa_2^{(4)2} + 16 \kappa_2^{(4)} \kappa_7^{(4)} + 8 \kappa_7^{(4)2}) - 48 \kappa_{15}^{(4)3} (9 \kappa_2^{(4)3} - 24 \kappa_2^{(4)2} \kappa_7^{(4)} - 23 \kappa_2^{(4)} \kappa_7^{(4)2} - 4 \kappa_7^{(4)3}) -$ $12 \kappa_{15}^{(4)} \kappa_2^{(4)} (18 \kappa_2^{(4)4} - 36 \kappa_2^{(4)3} \kappa_7^{(4)} - 9 \kappa_2^{(4)2} \kappa_7^{(4)2} - 6 \kappa_2^{(4)} \kappa_7^{(4)3} - 2 \kappa_7^{(4)4}) +$ $4 \kappa_{15}^{(4)2} (189 \kappa_2^{(4)4} + 180 \kappa_2^{(4)3} \kappa_7^{(4)} + 54 \kappa_2^{(4)2} \kappa_7^{(4)2} + 60 \kappa_2^{(4)} \kappa_7^{(4)3} + 4 \kappa_7^{(4)4}) +$ $36 \kappa_1^{(4)3} (36 \kappa_{15}^{(4)3} + 12 \kappa_{15}^{(4)2} (2 \kappa_2^{(4)} + \kappa_7^{(4)}) + 3 \kappa_2^{(4)} (12 \kappa_2^{(4)2} + \kappa_7^{(4)2}) - \kappa_{15}^{(4)} (24 \kappa_2^{(4)2} - 12 \kappa_2^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2})) +$ $9 \kappa_1^{(4)2} (432 \kappa_{15}^{(4)4} + 216 \kappa_2^{(4)4} + 36 \kappa_2^{(4)2} \kappa_7^{(4)2} + \kappa_7^{(4)4} + 48 \kappa_{15}^{(4)3} (5 \kappa_2^{(4)} + 4 \kappa_7^{(4)}) +$ $8 \kappa_{15}^{(4)2} (21 \kappa_2^{(4)2} + 22 \kappa_2^{(4)} \kappa_7^{(4)} + 5 \kappa_7^{(4)2}) + 4 \kappa_{15}^{(4)} (-36 \kappa_2^{(4)3} + 36 \kappa_2^{(4)2} \kappa_7^{(4)} + \kappa_2^{(4)} \kappa_7^{(4)2} + 2 \kappa_7^{(4)3}) +$ $6 \kappa_1^{(4)} (864 \kappa_{15}^{(4)5} + 144 \kappa_{15}^{(4)4} (7 \kappa_2^{(4)} + 4 \kappa_7^{(4)}) + 24 \kappa_{15}^{(4)3} (3 \kappa_2^{(4)2} + 20 \kappa_2^{(4)} \kappa_7^{(4)} + 9 \kappa_7^{(4)2}) +$ $8 \kappa_{15}^{(4)2} (36 \kappa_2^{(4)3} + 39 \kappa_2^{(4)2} \kappa_7^{(4)} + 12 \kappa_2^{(4)} \kappa_7^{(4)2} + 5 \kappa_7^{(4)3}) + 3 \kappa_2^{(4)} (72 \kappa_2^{(4)4} + 18 \kappa_2^{(4)2} \kappa_7^{(4)2} + \kappa_7^{(4)4}) +$ $\kappa_{15}^{(4)} (-144 \kappa_2^{(4)4} + 216 \kappa_2^{(4)3} \kappa_7^{(4)} + 30 \kappa_2^{(4)2} \kappa_7^{(4)2} + 24 \kappa_2^{(4)} \kappa_7^{(4)3} + 4 \kappa_7^{(4)4})))) /$ $(\kappa_{15}^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (-9 \kappa_1^{(4)} \kappa_{15}^{(4)} + 18 \kappa_{15}^{(4)2} + 6 \kappa_1^{(4)} \kappa_2^{(4)} - 15 \kappa_{15}^{(4)} \kappa_2^{(4)} + 6 \kappa_2^{(4)2} + 6 \kappa_{15}^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2}))$
Resolved unitarity condition(s):	(Demonstrably impossible)		